Chapter 9. The Fundamental Group

Path homotopy, covering spaces, the circle, retractions and fixed points, Borsuk–Ulam, homotopy type, spheres and surfaces

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9.1 Homotopy of paths

Motivation. Point-set topology cannot tell the plane from the punctured plane by counting components or checking compactness. What distinguishes them is a loop around the hole that cannot be shrunk to a point. Homotopy makes "deforming one path into another" precise, and it is the starting point of algebraic topology: turning shapes into groups.

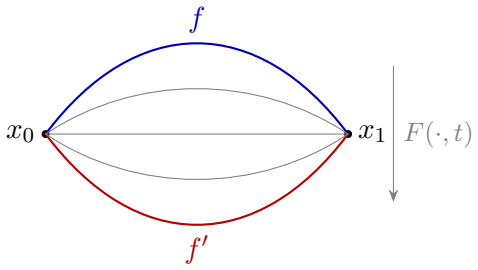


Figure. *A path homotopy F deforms f into f' through paths $F(\cdot,t)$ that all keep the end points x_0, x_1 fixed.*

Maps $f, f' : X \to Y$ are **homotopic**, $f \simeq f'$, if there is a continuous $F : X \times I \to Y$ ($I = [0,1]$) with $F(x,0) = f(x)$ and $F(x,1) = f'(x)$; f is **nulhomotopic** if it is homotopic to a constant map. Paths $f, f' : I \to X$ from x_0 to x_1 are **path homotopic**, $f \simeq_p f'$, if there is $F : I \times I \to X$ with $F(s,0) = f(s)$, $F(s,1) = f'(s)$, $F(0,t) = x_0$, $F(1,t) = x_1$.

Lemma 9.1 (51.1)

$\simeq$ and $\simeq_p$ are equivalence relations.

Proof. $F(x,t) = f(x)$ gives $f \simeq f$; if F is a homotopy from f to f', then $F(x,1-t)$ is one from f' to f. If $F : f \simeq f'$ and $F' : f' \simeq f''$, define $G(x,t) = F(x,2t)$ for $t \le \frac{1}{2}$ and $F'(x,2t-1)$ for $t \ge \frac{1}{2}$; the two formulas agree at $t = \frac{1}{2}$, so G is continuous by the pasting lemma. For paths, all these homotopies keep the end points fixed. ∎

We write $[f]$ for the path-homotopy class of f.

Example 9.1

If $f, g : X \to \mathbb{R}^2$ are maps (or paths with the same end points), the **straight-line homotopy** $F(x,t) = (1-t)f(x) + tg(x)$ is a homotopy (path homotopy). More generally this works for maps into any convex subset of $\mathbb{R}^n$. In $\mathbb{R}^2 - \{0\}$, however, the paths $f(s) = (\cos \pi s, \sin \pi s)$ and $g(s) = (\cos \pi s, -\sin \pi s)$ are not path homotopic (Section 9.4).

If f is a path from x_0 to x_1 and g a path from x_1 to x_2, the **product** $f * g$ is $f(2s)$ on $[0, \frac{1}{2}]$ and $g(2s-1)$ on $[\frac{1}{2}, 1]$. It is well defined on classes: if $F : f \simeq_p f'$ and $G : g \simeq_p g'$, then $H(s,t) = F(2s,t)$ for $s \le \frac{1}{2}$, $G(2s-1,t)$ for $s \ge \frac{1}{2}$, is a path homotopy $f * g \simeq_p f' * g'$. So $[f] * [g] = [f * g]$ is defined. Two observations will be used repeatedly: if $k : X \to Y$ is continuous and F is a path homotopy from f to f', then $k \circ F$ is a path homotopy from $k \circ f$ to $k \circ f'$; and $k \circ (f * g) = (k \circ f) * (k \circ g)$.

Theorem 9.2 (51.2)

The operation $*$ on path-homotopy classes satisfies:
(1) (associativity) if $[f] * ([g] * [h])$ is defined, so is $([f] * [g]) * [h]$, and they are equal;
(2) (identities) if e_x is the constant path at x and f runs from x_0 to x_1, then $[e_{x_0}] * [f] = [f] = [f] * [e_{x_1}]$;

(3) (inverses) with $\bar{f}(s) = f(1-s)$, $[f] * [\bar{f}] = [e_{x_0}]$ and $[\bar{f}] * [f] = [e_{x_1}]$.

Proof. Let e_0 be the constant path in I at 0, $i : I \to I$ the identity path, and $\bar{i}$ its reverse.

(2) $e_0 * i$ and i are paths in the convex set I from 0 to 1, so they are path homotopic by a straight-line homotopy G; then $f \circ G$ is a path homotopy between $f \circ (e_0 * i) = e_{x_0} * f$ and $f \circ i = f$. The other identity uses $i * e_1$.

(3) $i * \bar{i}$ and e_0 are paths in I from 0 to 0, hence path homotopic by some H; then $f \circ H$ shows $f * \bar{f} \simeq_p e_{x_0}$. Similarly $\bar{f} * f \simeq_p e_{x_1}$.

(1) For $0 < a < b < 1$ let $k_{a,b}$ be the path that on $[0, a]$, $[a, b]$, $[b, 1]$ is the positive linear map of that interval onto I followed by f, g, h respectively. Then $(f * g) * h = k_{1/2,3/4}$ and $f * (g * h) = k_{1/4,1/2}$. For any a, b, c, d, let $\gamma : I \to I$ be the piecewise linear map sending $0, a, b, 1$ to $0, c, d, 1$; then $k_{c,d} \circ \gamma = k_{a,b}$. Since γ and i are paths in I from 0 to 1, $\gamma \simeq_p i$, and composing with $k_{c,d}$ gives $k_{a,b} \simeq_p k_{c,d}$. ■

Theorem 9.3 (51.3)

Let f be a path in X and $0 = a_0 < a_1 < \cdots < a_n = 1$. Let f_i be the positive linear map of I onto $[a_{i-1}, a_i]$ followed by f. Then $[f] = [f_1] * \cdots * [f_n]$.

Proof. By associativity the right side is well defined. Represent it by a specific bracketing of $f_1 * \cdots * f_n$; this path is defined on a subdivision $0 = b_0 < \cdots < b_n = 1$ and on $[b_{i-1}, b_i]$ is the linear map onto I followed by f_i. Hence it equals $f \circ \gamma$, where $\gamma : I \to I$ is the piecewise linear map sending each $[b_{i-1}, b_i]$ linearly onto $[a_{i-1}, a_i]$. Since $\gamma \simeq_p i$ in I, $f \circ \gamma \simeq_p f$. ■

9.2 The fundamental group

Motivation. Loops at a base point, up to homotopy, form a group under concatenation. Homeomorphic spaces have isomorphic fundamental groups, so the group is an invariant that can prove two spaces are different.

For $x_0 \in X$, a **loop** at x_0 is a path from x_0 to x_0. By Theorem 9.2 the path-homotopy classes of loops at x_0 form a group under $*$, the **fundamental group** $\pi_1(X, x_0)$. If X is a convex subset of $\mathbb{R}^n$, $\pi_1(X, x_0)$ is trivial (straight-line homotopy to the constant loop). For a path α from x_0 to x_1 define $\hat{\alpha} : \pi_1(X, x_0) \to \pi_1(X, x_1)$ by $\hat{\alpha}([f]) = [\bar{\alpha}] * [f] * [\alpha]$.

Theorem 9.4 (52.1)

$\hat{\alpha}$ is a group isomorphism.

Proof. $\hat{\alpha}([f]) * \hat{\alpha}([g]) = [\bar{\alpha}] * [f] * [\alpha] * [\bar{\alpha}] * [g] * [\alpha] = \hat{\alpha}([f] * [g])$. With $\beta = \bar{\alpha}$, $\hat{\beta}(\hat{\alpha}([h])) = [\alpha] * [\bar{\alpha}] * [h] * [\alpha] * [\bar{\alpha}] = [h]$, and similarly $\hat{\alpha} \circ \hat{\beta} = \mathrm{id}$. ■

Corollary 9.5 (52.2)

If X is path connected and $x_0, x_1 \in X$, then $\pi_1(X, x_0) \cong \pi_1(X, x_1)$.

If C is the path component of x_0, then $\pi_1(C, x_0) = \pi_1(X, x_0)$, since loops and homotopies based at x_0 lie in C. A path-connected space with trivial fundamental group is **simply connected**.

Lemma 9.6 (52.3)

In a simply connected space, any two paths with the same initial and final points are path homotopic.

Proof. If f, g run from x_0 to x_1, then $f * \bar{g}$ is a loop at x_0, so $[f] * [\bar{g}] = [e_{x_0}]$; multiplying on the right by $[g]$ gives $[f] = [g]$. ■

A continuous $h : (X, x_0) \to (Y, y_0)$ (that is, $h(x_0) = y_0$) induces $h_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$, $h_*([f]) = [h \circ f]$, well defined and a homomorphism by the observations before Theorem 9.2.

Theorem 9.7 (52.4)

If $h : (X, x_0) \to (Y, y_0)$ and $k : (Y, y_0) \to (Z, z_0)$ are continuous, then $(k \circ h)_* = k_* \circ h_*$. If i is the identity of (X, x_0), then i_* is the identity.

Proof. $(k \circ h)_*([f]) = [(k \circ h) \circ f] = [k \circ (h \circ f)] = k_*(h_*([f]))$, and $i_*([f]) = [f]$. ■

Corollary 9.8 (52.5)

If $h : (X, x_0) \to (Y, y_0)$ is a homeomorphism, h_* is an isomorphism.

Proof. With $k = h^{-1}$, $k_* h_* = (k \circ h)_* = \mathrm{id}$ and $h_* k_* = \mathrm{id}$. ■

9.3 Covering spaces

Motivation. To compute a fundamental group we need a way to tell loops apart. A covering space unwraps the space so that loops lift to paths, and the endpoint of the lifted path detects how the loop winds; $\mathbb{R} \to S^1$, $t \mapsto e^{2\pi i t}$ is the model example.

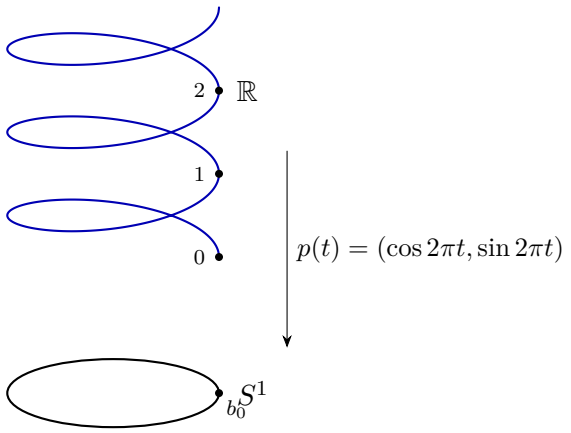


Figure. *The covering map $p : \mathbb{R} \to S^1$ wraps the line around the circle; the fibre $p^{-1}(b_0) = \mathbb{Z}$. A loop at b_0 lifts to a path from 0 to an integer, and that integer is the loop's class in $\pi_1(S^1) \cong \mathbb{Z}$.*

Let $p : E \to B$ be continuous and surjective. An open $U \subset B$ is **evenly covered** by p if $p^{-1}(U)$ is a union of disjoint open sets V_α (the **slices**), each mapped homeomorphically onto U by p. If every point of B has an evenly covered neighbourhood, p is a **covering map** and E a **covering space** of B. A covering map is a local homeomorphism and an open map, and each fibre $p^{-1}(b)$ is discrete (each of its points is the only one in its slice).

Theorem 9.9 (53.1)

$p : \mathbb{R} \to S^1$, $p(x) = (\cos 2\pi x, \sin 2\pi x)$, is a covering map.

Proof. Let U be the open arc of points with positive first coordinate. Then $p^{-1}(U) = \bigcup_n (n - \frac{1}{4}, n + \frac{1}{4})$. On $[n - \frac{1}{4}, n + \frac{1}{4}]$, p is injective (its second coordinate is strictly increasing), so it maps this compact interval homeomorphically onto the closed right half circle; restricting, $(n - \frac{1}{4}, n + \frac{1}{4})$ maps homeomorphically onto U. The same argument applies to the arcs with negative first coordinate, positive second coordinate and negative second coordinate, and these four arcs cover S^1. ■

Example 9.2

The restriction of p to $(0, \infty)$ is a surjective local homeomorphism but not a covering map: no neighbourhood of $b_0 = (1, 0)$ is evenly covered, because the component of $p^{-1}(U)$ containing a point near 0 is a half-open interval $(0, \epsilon)$ that is not mapped onto U.

Theorem 9.10 (53.2)

If $p : E \to B$ is a covering map, $B_0 \subset B$ and $E_0 = p^{-1}(B_0)$, then $p_0 = p|E_0 : E_0 \to B_0$ is a covering map.

Proof. For $b \in B_0$ let U be evenly covered with slices V_α. Then $U \cap B_0$ is a neighbourhood of b in B_0, and $p_0^{-1}(U \cap B_0)$ is the disjoint union of the sets $V_\alpha \cap E_0$, open in E_0, each mapped homeomorphically onto $U \cap B_0$. ■

Theorem 9.11 (53.3)

If $p : E \to B$ and $p' : E' \to B'$ are covering maps, so is $p \times p' : E \times E' \to B \times B'$.

Proof. If U, U' are evenly covered neighbourhoods of b, b' with slices V_α, V'_β, then $(p \times p')^{-1}(U \times U')$ is the disjoint union of the open sets $V_\alpha \times V'_\beta$, each mapped homeomorphically onto $U \times U'$. ■

Example 9.3

$p \times p : \mathbb{R} \times \mathbb{R} \to S^1 \times S^1$ is a covering of the torus $T = S^1 \times S^1$. Restricting to $b_0 \times S^1 \cup S^1 \times b_0$ (Theorem 9.10) gives a covering of the figure eight $b_0 \times S^1 \cup S^1 \times b_0$ by the grid of horizontal and vertical lines through integer points. Since $\mathbb{R}_+ \times S^1 \cong \mathbb{R}^2 - 0$ by polar coordinates, $\mathrm{id} \times p : \mathbb{R}_+ \times \mathbb{R} \to \mathbb{R}^2 - 0$ is a covering of the punctured plane.

9.4 The fundamental group of the circle

Motivation. Using the covering $\mathbb{R} \to S^1$ we prove $\pi_1(S^1) \cong \mathbb{Z}$, the first nontrivial computation. The rest of the chapter is a series of applications of this one result.

If $p : E \to B$ is a map, a **lifting** of $f : X \to B$ is a map $\tilde{f} : X \to E$ with $p \circ \tilde{f} = f$.

Lemma 9.12 (54.1)

Let $p : E \to B$ be a covering map with $p(e_0) = b_0$. Any path $f : I \to B$ beginning at b_0 has a unique lifting $\tilde{f}$ beginning at e_0.

Proof. Cover B by evenly covered open sets; by the Lebesgue number lemma there is a subdivision $0 = s_0 < \cdots < s_n = 1$ such that each $f([s_i, s_{i+1}])$ lies in an evenly covered U.

Existence: set $\tilde{f}(0) = e_0$; if $\tilde{f}$ is defined on $[0, s_i]$, let V_0 be the slice of $p^{-1}(U)$ containing $\tilde{f}(s_i)$ and define $\tilde{f}(s) = (p|V_0)^{-1}(f(s))$ for $s \in [s_i, s_{i+1}]$; the pasting lemma gives continuity.

Uniqueness: if $\tilde{f}'$ is another lifting and $\tilde{f}' = \tilde{f}$ on $[0, s_i]$, then $\tilde{f}'([s_i, s_{i+1}])$ is connected and lies in the disjoint union of the open slices, hence in one slice, which must be V_0 since it contains $\tilde{f}'(s_i) = \tilde{f}(s_i)$. Then $\tilde{f}'(s) = (p|V_0)^{-1}(f(s)) = \tilde{f}(s)$ there. ■

Lemma 9.13 (54.2)

Let $p : E \to B$ be a covering map with $p(e_0) = b_0$, and $F : I \times I \to B$ continuous with $F(0,0) = b_0$. There is a unique lifting $\tilde{F} : I \times I \to E$ with $\tilde{F}(0,0) = e_0$. If F is a path homotopy, so is $\tilde{F}$.

Proof. Existence: choose subdivisions $s_0 < \cdots < s_m$ and $t_0 < \cdots < t_n$ of I such that each rectangle $I_i \times J_j = [s_{i-1}, s_i] \times [t_{j-1}, t_j]$ is mapped by F into an evenly covered open set (Lebesgue number lemma). Define $\tilde{F}$ on $A = 0 \times I \cup I \times 0$ by lifting the two edges as paths from $(0,0)$ (Lemma 9.12). Then extend over the rectangles in the order $I_1 \times J_1, \ldots, I_m \times J_1, I_1 \times J_2, \ldots$. At each stage let A' be A together with the rectangles already treated, R the next rectangle and $C = R \cap A'$, the union of the left and bottom edges of R, which is connected. With $F(R) \subset U$ evenly covered, $\tilde{F}(C)$ is connected, so lies in one slice V_0; define $\tilde{F} = (p|V_0)^{-1} \circ F$ on R. This agrees with the earlier definition on C and is continuous on $A' \cup R$ by the pasting lemma. Uniqueness: any lifting is determined on A by Lemma 9.12, and on each successive R its image is connected, contains $\tilde{F}(C) \subset V_0$, hence lies in V_0, so it equals $(p|V_0)^{-1} \circ F$. If F is a path homotopy, $F(0 \times I) = b_0$, so $\tilde{F}(0 \times I)$ is a connected subset of the discrete fibre $p^{-1}(b_0)$, hence a point; likewise for $1 \times I$. ■

Theorem 9.14 (54.3)

Let $p : E \to B$ be a covering map, $p(e_0) = b_0$, and f, g paths in B from b_0 to b_1 with liftings $\tilde{f}, \tilde{g}$ beginning at e_0. If $f \simeq_p g$, then $\tilde{f}$ and $\tilde{g}$ end at the same point and $\tilde{f} \simeq_p \tilde{g}$.

Proof. Let $F : f \simeq_p g$ and $\tilde{F}$ its lifting with $\tilde{F}(0,0) = e_0$, a path homotopy (Lemma 9.13). $\tilde{F}|I \times 0$ lifts f and begins at e_0, so equals $\tilde{f}$; $\tilde{F}|I \times 1$ lifts g and begins at $\tilde{F}(0,1) = e_0$, so equals $\tilde{g}$. Since $\tilde{F}(1 \times I)$ is a point, $\tilde{f}(1) = \tilde{g}(1)$. ■

The **lifting correspondence** $\phi : \pi_1(B, b_0) \to p^{-1}(b_0)$ is $\phi([f]) = \tilde{f}(1)$, where $\tilde{f}$ lifts f from e_0; it is well defined by Theorem 9.14.

Theorem 9.15 (54.4)

If E is path connected, ϕ is surjective; if E is simply connected, ϕ is bijective.

Proof. For $e_1 \in p^{-1}(b_0)$ take a path γ in E from e_0 to e_1; γ lifts the loop $f = p \circ \gamma$, so $\phi([f]) = e_1$. If $\phi([f]) = \phi([g])$ and E is simply connected, then $\tilde{f} \simeq_p \tilde{g}$ by Lemma 9.6, say by $\tilde{F}$, and $p \circ \tilde{F}$ is a path homotopy from f to g. ■

Theorem 9.16 (54.5)

$\pi_1(S^1, b_0)$ is isomorphic to the additive group $\mathbb{Z}$.

Proof. Use $p : \mathbb{R} \to S^1$ with $e_0 = 0$; $p^{-1}(b_0) = \mathbb{Z}$ and $\mathbb{R}$ is simply connected, so ϕ is a bijection onto $\mathbb{Z}$. If $\tilde{f}, \tilde{g}$ lift f, g from 0 with $\tilde{f}(1) = n$, $\tilde{g}(1) = m$, then $s \mapsto n + \tilde{g}(s)$ lifts g from n (as $p(n+x) = p(x)$),

so $\tilde{f} * (n + \tilde{g})$ lifts $f * g$ from 0 and ends at $n + m$. Thus $\phi([f] * [g]) = \phi([f]) + \phi([g])$. ■

In particular the loop $p_0(s) = p(s)$ generates $\pi_1(S^1, b_0)$, and the class of a loop is determined by the end point n of its lifting, the number of times it winds around.

Theorem 9.17 (54.6)

Let $p : E \to B$ be a covering map, $p(e_0) = b_0$.
(a) $p_* : \pi_1(E, e_0) \to \pi_1(B, b_0)$ is injective.
(b) Let $H = p_*(\pi_1(E, e_0))$. The lifting correspondence induces an injective map $\Phi : \pi_1(B, b_0)/H \to p^{-1}(b_0)$ of the right cosets of H, bijective if E is path connected.
(c) A loop f at b_0 has $[f] \in H$ iff f lifts to a loop at e_0.

Proof. (a) Let $\tilde{h}$ be a loop at e_0 with $p \circ \tilde{h} \simeq_p e_{b_0}$ by F. The lifting $\tilde{F}$ from e_0 is a path homotopy between the liftings of $p \circ \tilde{h}$ and e_{b_0} from e_0, namely $\tilde{h}$ and the constant loop.

(c) If $[f] = [p \circ \tilde{h}]$ with $\tilde{h}$ a loop at e_0, then by Theorem 9.14 $\tilde{f}$ ends where $\tilde{h}$ does, at e_0. Conversely, if $\tilde{f}$ is a loop, $[f] = p_*[\tilde{f}]$.

(b) Define $\Phi(H * [f]) = \phi([f])$. If $[h] \in H$, $h = p \circ \tilde{h}$ with $\tilde{h}$ a loop at e_0, then $\tilde{h} * \tilde{f}$ lifts $h * f$ and ends at $\tilde{f}(1)$; so Φ is well defined. If $\phi([f]) = \phi([g])$, then $\tilde{f} * \bar{\tilde{g}}$ is a loop at e_0 lifting $f * \bar{g}$, so $[f] * [g]^{-1} \in H$ and $H * [f] = H * [g]$. Surjectivity for path-connected E is Theorem 9.15. ■

9.5 Retractions and fixed points

Motivation. Since $\pi_1(S^1) \neq 0$ while the disc is simply connected, there is no retraction of the disc onto its boundary. This gives the Brouwer fixed-point theorem in dimension two and a criterion for when a vector field must vanish.

$A \subset X$ is a **retract** of X if there is a continuous $r : X \to A$ with $r|A = \mathrm{id}_A$ (a **retraction**).

Lemma 9.18 (55.1)

If A is a retract of X, the homomorphism j_* induced by inclusion $j : A \to X$ is injective.

Proof. $r \circ j = \mathrm{id}_A$, so $r_* \circ j_*$ is the identity. ■

Theorem 9.19 (55.2, no-retraction theorem)

There is no retraction of B^2 onto S^1.

Proof. $j_* : \pi_1(S^1) \cong \mathbb{Z} \to \pi_1(B^2) = 0$ is not injective. ■

Lemma 9.20 (55.3)

For continuous $h : S^1 \to X$ the following are equivalent:
(1) h is nulhomotopic;
(2) h extends to a continuous $k : B^2 \to X$;
(3) h_* is the trivial homomorphism.

Proof. (1)⇒(2): let $H : S^1 \times I \to X$ be a homotopy from h to a constant. The map $\pi : S^1 \times I \to B^2$, $\pi(x, t) = (1 - t)x$, is continuous, closed and surjective, hence a quotient map; it collapses $S^1 \times 1$ to 0 and is otherwise injective. H is constant on $S^1 \times 1$, so it induces a continuous k with $k \circ \pi = H$, and

k extends h. (2)⇒(3): $h = k \circ j$ with $j : S^1 \to B^2$, and j_* is trivial. (3)⇒(1): let $p_0(s) = p(s)$ be the standard loop and $b_0 = p_0(0)$. Then $h \circ p_0$ is path homotopic to a constant by some $F : I \times I \to X$, with $F(0,t) = F(1,t) = h(b_0)$ and $F(s,1) = h(b_0)$. The map $q : I \times I \to B^2$, $q(s,t) = (1-t)p_0(s)$, is a quotient map that identifies $(0,t)$ with $(1,t)$ and collapses $I \times 1$; F respects these identifications, so it induces $k : B^2 \to X$ with $k \circ q = F$, and k extends h. Then $H(x,t) = k((1-t)x)$ is a homotopy from h to the constant $k(0)$. ■

Corollary 9.21 (55.4)

The inclusion $j : S^1 \to \mathbb{R}^2 - 0$ and the identity $i : S^1 \to S^1$ are not nulhomotopic.

Proof. $r(x) = x/\|x\|$ retracts $\mathbb{R}^2 - 0$ onto S^1, so j_* is injective (Lemma 9.18), hence nontrivial; i_* is the identity of $\mathbb{Z}$. Apply Lemma 9.20. ■

Theorem 9.22 (55.5)

Given a nonvanishing vector field on B^2 (a continuous $v : B^2 \to \mathbb{R}^2 - 0$), there is a point of S^1 where v points directly inward and one where it points directly outward.

Proof. Suppose $v(x)$ never points directly inward, i.e. $v(x) \neq ax$ for $x \in S^1$, $a < 0$. Then $w = v|S^1 : S^1 \to \mathbb{R}^2 - 0$ extends to B^2, so is nulhomotopic. $F(x,t) = tx + (1-t)w(x)$ never vanishes on $S^1 \times I$: for $t = 0, 1$ clearly, and for $0 < t < 1$, $F = 0$ would mean $w(x) = -\frac{t}{1-t}x$. So $j \simeq w$ in $\mathbb{R}^2 - 0$ and j is nulhomotopic, contradicting Corollary 9.21. For outward, apply this to $-v$. ■

Theorem 9.23 (55.6, Brouwer fixed-point theorem for the disc)

Every continuous $f : B^2 \to B^2$ has a fixed point.

Proof. Otherwise $v(x) = f(x) - x$ is a nonvanishing vector field on B^2. At $x \in S^1$ it cannot point directly outward, since $f(x) - x = ax$ with $a > 0$ gives $f(x) = (1+a)x \notin B^2$. This contradicts Theorem 9.22. ■

Corollary 9.24 (55.7)

A 3×3 matrix A of positive real numbers has a positive real eigenvalue.

Proof. Let $B = \{x \in S^2 : x_1, x_2, x_3 \geq 0\}$; radial projection $x \mapsto x/(x_1+x_2+x_3)$ is a homeomorphism of B onto a closed triangle, which is homeomorphic to B^2. For $x \in B$, Ax has positive entries, so $x \mapsto Ax/\|Ax\|$ maps B into B. A fixed point gives $Ax = \|Ax\|x$. ■

Theorem 9.25 (55.8)

Let T be a closed triangle in $\mathbb{R}^2$. There is $\epsilon > 0$ such that for every open covering $\mathcal{A}$ of T by sets of diameter less than ϵ, some point of T belongs to at least three elements of $\mathcal{A}$.

Proof. Since no point lies on all three edges e_1, e_2, e_3, the continuous function $(a,b,c) \mapsto \max(|a-b|, |b-c|, |a-c|)$ on the compact $e_1 \times e_2 \times e_3$ is positive and has a minimum $\epsilon > 0$; so no set of diameter less than ϵ meets all three edges. Suppose $\mathcal{A}$ is such a covering in which no point lies in three elements; passing to a finite subcovering, take a dominated partition of unity $\{\phi_A\}$. For each A choose a vertex v_A such that A does not meet the edge opposite v_A. Define $k(x) = \sum_A \phi_A(x) v_A$. Each x lies in at most two elements, so $k(x)$ is a convex combination of at most two vertices and lies in ∂T. If x lies on an edge e and $\phi_A(x) > 0$, then A meets e, so e is not opposite v_A, i.e. $v_A \in e$; thus

$k(e) \subset e$. Then $H(x,t) = (1-t)x + tk(x)$ is a homotopy in ∂T from the identity of ∂T to $k|\partial T$ (it stays in an edge containing x). But $k|\partial T$ extends to T, so is nulhomotopic (Lemma 9.20, $T \cong B^2$, $\partial T \cong S^1$), and then the identity of $\partial T \cong S^1$ would be nulhomotopic, contradicting Corollary 9.21. ■

So the triangle has topological dimension at least 2.

9.6 The fundamental theorem of algebra

Motivation. A topological proof: a polynomial of degree n wraps a large circle n times around the origin, and if it had no root this loop would be contractible in $\mathbb{C} - \{0\}$, contradicting $\pi_1(S^1) \cong \mathbb{Z}$.

Theorem 9.26 (56.1, fundamental theorem of algebra)

A polynomial equation $x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0$ of degree $n > 0$ with real or complex coefficients has a complex root.

Proof.

Step 1. $f : S^1 \to S^1$, $f(z) = z^n$, induces an injective homomorphism: $f \circ p_0(s) = p(ns)$ lifts to $s \mapsto ns$, so f_* is multiplication by n on $\mathbb{Z}$.

Step 2. $g : S^1 \to \mathbb{R}^2 - 0$, $g(z) = z^n$, equals $j \circ f$, so $g_* = j_* f_*$ is injective and g is not nulhomotopic.

Step 3. if $\sum|a_i| < 1$, there is a root in B^2. Otherwise $k(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0$ maps B^2 into $\mathbb{R}^2 - 0$, so $h = k|S^1$ is nulhomotopic. But $F(z,t) = z^n + t(a_{n-1}z^{n-1} + \cdots + a_0)$ never vanishes on $S^1 \times I$, since $|t(a_{n-1}z^{n-1} + \cdots + a_0)| \le \sum|a_i| < 1 = |z^n|$; so $g \simeq h$ in $\mathbb{R}^2 - 0$, a contradiction.

Step 4. in general, choose $c > 0$ with $\sum_i |a_{n-i}|/c^i < 1$ and substitute $x = cy$: the equation $y^n + (a_{n-1}/c)y^{n-1} + \cdots + a_0/c^n = 0$ has a root y_0, and cy_0 is a root of the original. ■

9.7 The Borsuk–Ulam theorem

Motivation. Some point on the earth has the same temperature and pressure as its antipode. The Borsuk–Ulam theorem for S^2 proves this, again using the fact that a nontrivial loop in S^1 cannot be killed.

$h : S^n \to S^m$ is **antipode-preserving** if $h(-x) = -h(x)$.

Theorem 9.27 (57.1)

A continuous antipode-preserving $h : S^1 \to S^1$ is not nulhomotopic.

Proof. Composing with a rotation (which commutes with $x \mapsto -x$) we may assume $h(b_0) = b_0$. Regard $S^1 \subset \mathbb{C}$ and let $q(z) = z^2$, a quotient map (continuous, closed, surjective) identifying z with $-z$. Since $q(h(-z)) = q(-h(z)) = q(h(z))$, there is a continuous $k : S^1 \to S^1$ with $k \circ q = q \circ h$. Now q_* is injective ($q \circ p_0$ lifts to $s \mapsto 2s$, so q_* is multiplication by 2). Let $f(s) = e^{\pi i s}$, a path from b_0 to $-b_0$; then $q \circ f = p_0$. The path $h \circ f$ runs from b_0 to $-b_0$, so its lifting $\tilde{g}$ from 0 ends at some $m + \frac{1}{2}$; $q \circ h \circ f = p \circ 2\tilde{g}$, and $2\tilde{g}$ ends at the odd integer $2m+1$. So $k_*[p_0] = [k \circ q \circ f] = [q \circ h \circ f]$ corresponds to $2m + 1 \neq 0$, and k_* is injective. If h were nulhomotopic, h_* would be trivial, hence $k_* q_* = q_* h_*$ trivial, contradicting the injectivity of $k_* q_*$. ■

Theorem 9.28 (57.2)

There is no continuous antipode-preserving map $g : S^2 \to S^1$.

Proof. Its restriction h to the equator S^1 is antipode-preserving, hence not nulhomotopic. But h extends to the upper hemisphere, which is homeomorphic to B^2 by $(x, y, z) \mapsto (x, y)$; by Lemma 9.20 h is nulhomotopic. ■

Theorem 9.29 (57.3, Borsuk–Ulam theorem for S^2)

For every continuous $f : S^2 \to \mathbb{R}^2$ there is $x \in S^2$ with $f(x) = f(-x)$.

Proof. Otherwise $g(x) = (f(x) - f(-x))/\|f(x) - f(-x)\|$ is a continuous antipode-preserving map $S^2 \to S^1$. ■

Theorem 9.30 (57.4, bisection theorem)

Given two bounded polygonal regions in $\mathbb{R}^2$, some line bisects each of them.

Proof. Put the regions A_1, A_2 in the plane $z = 1$ of $\mathbb{R}^3$. For $u \in S^2$, let $f_i(u)$ be the area of the part of A_i on the side $x \cdot u > 0$ of the plane $P_u = \{x \cdot u = 0\}$. Then $f_i(-u) = \text{area}(A_i) - f_i(u)$, and f_i is continuous (we assume this elementary fact about areas). Borsuk–Ulam applied to (f_1, f_2) gives u with $f_i(u) = f_i(-u) = \frac{1}{2}\,\text{area}(A_i)$. Here $u \neq \pm(0, 0, 1)$, since then $f_i(u)$ is 0 or the full area; so P_u meets the plane $z = 1$ in a line, which bisects both regions. ■

9.8 Deformation retracts and homotopy type

Motivation. The punctured plane and the circle have the same fundamental group because one can be squeezed onto the other. Homotopy equivalence makes this idea precise and lets us replace a space by a simpler one before computing.

Lemma 9.31 (58.1)

If $h, k : (X, x_0) \to (Y, y_0)$ are homotopic by a homotopy keeping x_0 at y_0, then $h_* = k_*$.

Proof. For a loop f at x_0, $G(s, t) = H(f(s), t)$ is a path homotopy from $h \circ f$ to $k \circ f$. ■

Theorem 9.32 (58.2)

The inclusion $j : S^n \to \mathbb{R}^{n+1} - 0$ induces an isomorphism of fundamental groups.

Proof. $r(x) = x/\|x\|$ satisfies $r \circ j = \text{id}$, and $H(x, t) = (1 - t)x + tx/\|x\|$ is a homotopy in $\mathbb{R}^{n+1} - 0$ from the identity to $j \circ r$ fixing S^n pointwise. By Lemma 9.31, $j_* r_* = \text{id}$, and $r_* j_* = \text{id}$. ■

$A \subset X$ is a **deformation retract** of X if the identity of X is homotopic to a map carrying X into A by a homotopy fixing each point of A: $H : X \times I \to X$ with $H(x, 0) = x$, $H(x, 1) \in A$, $H(a, t) = a$ for $a \in A$.

Theorem 9.33 (58.3)

If A is a deformation retract of X and $x_0 \in A$, then $j : (A, x_0) \to (X, x_0)$ induces an isomorphism.

Proof. As in Theorem 9.32, with $r(x) = H(x, 1)$. ∎

Example 9.4

The doubly punctured plane $\mathbb{R}^2 - \{p, q\}$ deformation retracts onto a figure eight, and also onto a theta space; the solid torus $B^2 \times S^1$ and $\mathbb{R}^3$ minus the z-axis deformation retract onto a circle, so their fundamental groups are $\mathbb{Z}$.

$f : X \to Y$ is a **homotopy equivalence** with **homotopy inverse** $g : Y \to X$ if $g \circ f \simeq \mathrm{id}_X$ and $f \circ g \simeq \mathrm{id}_Y$; then X, Y have the same **homotopy type**.

Lemma 9.34 (58.4)

Let $h, k : X \to Y$ be homotopic by H, with $h(x_0) = y_0$, $k(x_0) = y_1$, and let $\alpha(t) = H(x_0, t)$. Then $k_* = \hat{\alpha} \circ h_*$.

Proof. Let f be a loop at x_0 and $G(s, t) = H(f(s), t)$ on $I \times I$. Consider the paths $\beta_0(s) = (s, 0)$, $\beta_1(s) = (s, 1)$, $\gamma_0(t) = (0, t)$, $\gamma_1(t) = (1, t)$ in $I \times I$. Both $\beta_0 * \gamma_1$ and $\gamma_0 * \beta_1$ run from $(0, 0)$ to $(1, 1)$ in the convex set $I \times I$, hence are path homotopic, and so are their images under G: $(h \circ f) * \alpha \simeq_p \alpha * (k \circ f)$. Thus $[k \circ f] = [\bar{\alpha}] * [h \circ f] * [\alpha]$. ∎

Corollary 9.35 (58.5)

If $h \simeq k$ and h_* is injective, surjective or trivial, so is k_*.

Corollary 9.36 (58.6)

If h is nulhomotopic, h_* is trivial.

Proof. Both follow from Lemma 9.34, since $\hat{\alpha}$ is an isomorphism and a constant map induces the trivial homomorphism. ∎

Theorem 9.37 (58.7)

If $f : X \to Y$ is a homotopy equivalence with $f(x_0) = y_0$, then $f_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$ is an isomorphism.

Proof. Let g be a homotopy inverse, $x_1 = g(y_0)$, $y_1 = f(x_1)$. Since $g \circ f \simeq \mathrm{id}_X$, Lemma 9.34 shows that $g_* \circ (f_{x_0})_* : \pi_1(X, x_0) \to \pi_1(X, x_1)$ is an isomorphism, so $(g_{y_0})_*$ is surjective. Since $f \circ g \simeq \mathrm{id}_Y$, $(f_{x_1})_* \circ (g_{y_0})_* : \pi_1(Y, y_0) \to \pi_1(Y, y_1)$ is an isomorphism, so $(g_{y_0})_*$ is injective. Hence $(g_{y_0})_*$ is an isomorphism, and so is $(f_{x_0})_*$. ∎

9.9 The fundamental group of S^n

Motivation. Higher spheres are simply connected. The proof introduces the idea, made systematic by van Kampen's theorem in Chapter 11, of computing π_1 from a cover by open sets with known fundamental groups.

Theorem 9.38 (59.1)

Let $X = U \cup V$ with U, V open, $U \cap V$ path connected and $x_0 \in U \cap V$. Then the images of $i_* : \pi_1(U, x_0) \to \pi_1(X, x_0)$ and $j_* : \pi_1(V, x_0) \to \pi_1(X, x_0)$ generate $\pi_1(X, x_0)$.

Proof. Let f be a loop at x_0. By the Lebesgue number lemma choose $0 = a_0 < \cdots < a_n = 1$ with each $f([a_{i-1}, a_i])$ in U or in V. If some $f(a_i)$ $(0 < i < n)$ is not in $U \cap V$, say it lies in $U - V$, then both adjacent intervals are mapped into U, and a_i can be deleted; repeating, we may assume $f(a_i) \in U \cap V$ for all i. Choose paths α_i in $U \cap V$ from x_0 to $f(a_i)$ (α_0, α_n constant). With f_i as in Theorem 9.3, $[f] = [f_1] * \cdots * [f_n] = [g_1] * \cdots * [g_n]$ where $g_i = \alpha_{i-1} * f_i * \bar{\alpha}_i$ is a loop at x_0 lying entirely in U or in V. ∎

Corollary 9.39 (59.2)

If $X = U \cup V$ with U, V open and simply connected and $U \cap V$ nonempty and path connected, then X is simply connected.

Theorem 9.40 (59.3)

For $n \geq 2$, S^n is simply connected.

Proof. Let $p = (0, \ldots, 0, 1)$. Stereographic projection $f(x) = (x_1, \ldots, x_n)/(1 - x_{n+1})$ is a homeomorphism $S^n - p \to \mathbb{R}^n$, with inverse $y \mapsto (t(y)y, 1 - t(y))$, $t(y) = 2/(1 + \|y\|^2)$; similarly $S^n - (-p) \cong \mathbb{R}^n$. Both are simply connected, and their intersection is homeomorphic to $\mathbb{R}^n - 0$, which is path connected for $n \geq 2$. Apply Corollary 9.39. ∎

9.10 Fundamental groups of some surfaces

Motivation. The torus, the projective plane and the double torus can now be distinguished from each other and from the sphere by their fundamental groups, giving the first steps toward the classification of surfaces.

A **surface** is a Hausdorff space with a countable basis in which each point has a neighbourhood homeomorphic to an open subset of $\mathbb{R}^2$.

Theorem 9.41 (60.1)

$\pi_1(X \times Y, x_0 \times y_0) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0)$.

Proof. With projections p, q, let $\Phi([f]) = (p_*[f], q_*[f])$, a homomorphism. Given loops g in X and h in Y, $f(s) = g(s) \times h(s)$ maps to $([g], [h])$. If $p \circ f \simeq_p e$ by G and $q \circ f \simeq_p e$ by H, then $G \times H$ is a path homotopy from f to the constant loop. ∎

Corollary 9.42 (60.2)

The fundamental group of the torus $S^1 \times S^1$ is $\mathbb{Z} \times \mathbb{Z}$.

The **projective plane** P^2 is the quotient of S^2 obtained by identifying each x with $-x$.

Theorem 9.43 (60.3)

P^2 is a compact surface, and the quotient map $p : S^2 \to P^2$ is a covering map.

Proof. p is open: for U open, $p^{-1}(p(U)) = U \cup a(U)$ with $a(x) = -x$, which is open. For $x \in S^2$ let U be the ϵ-neighbourhood of x with $\epsilon < 1$; it contains no antipodal pair, so $p|U$ is a continuous open bijection onto $p(U)$, a homeomorphism, and similarly for $a(U)$, which is disjoint from U. So $p(U)$ is evenly covered with slices $U, a(U)$. Hausdorff: for $p(x) \neq p(y)$, the four points $\pm x, \pm y$ are distinct; take ϵ less than half their minimum distance, so the ϵ-neighbourhoods U of x and W of y satisfy $(U \cup a(U)) \cap (W \cup a(W)) = \emptyset$, whence $p(U) \cap p(W) = \emptyset$. A countable basis of S^2 maps to one of P^2 since p is open. Locally, P^2 is homeomorphic to open subsets of S^2, hence of $\mathbb{R}^2$; and it is compact as a continuous image of S^2. ■

Corollary 9.44 (60.4)

$\pi_1(P^2, y)$ has order 2.

Proof. S^2 is simply connected, so the lifting correspondence is a bijection onto the two-point fibre (Theorem 9.15). ■

Similarly P^n $(n \geq 2)$ has fundamental group of order 2.

Lemma 9.45 (60.5)

The fundamental group of the figure eight is not abelian.

Proof. Let $X = A \cup B$ be two circles meeting at x_0. Let E be the union of the coordinate axes in $\mathbb{R}^2$ together with, for each nonzero integer n, a circle tangent to the x-axis at $(n, 0)$ and a circle tangent to the y-axis at $(0, n)$. Define $p : E \to X$ by wrapping the x-axis around A (each $[n, n+1]$ once, integers to x_0), the y-axis around B, each circle tangent to the x-axis homeomorphically onto B and each circle tangent to the y-axis onto A (tangency points to x_0). Points of $X - x_0$ have small arc neighbourhoods whose preimages are disjoint arcs mapped homeomorphically, and x_0 has a small "cross" neighbourhood whose preimage is a disjoint union of crosses about the integer lattice points on the axes; so p is a covering map. Let g and h be loops going once around A and B. The lifting of $g * h$ from the origin runs along the x-axis to $(1, 0)$ and then around the circle tangent there, ending at $(1, 0)$; the lifting of $h * g$ ends at $(0, 1)$. By Theorem 9.14, $g * h \not\simeq_p h * g$. ■

Theorem 9.46 (60.6)

The fundamental group of the double torus is not abelian.

Proof. The double torus X contains a figure eight A, consisting of one circle around each handle meeting at a point, and A is a retract of X: each handle can be squashed continuously onto its circle, compatibly along the curve where the two halves are joined. By Lemma 9.18, $\pi_1(X)$ contains a copy of the nonabelian group $\pi_1(A)$. ■

Corollary 9.47 (60.7)

The 2-sphere, torus, projective plane and double torus are topologically distinct.

Proof. Their fundamental groups are trivial, $\mathbb{Z} \times \mathbb{Z}$ (infinite abelian), of order 2, and nonabelian. ■